



CS-404 Big Data Analytics ASSIGNMENT 02

Data Source Selection & Ingestion Pipeline

BDA Spring 2026 By: Ms. Zahida Kausar (Total Marks: 30)

Submission deadline: Wednesday 15th April, 2026 (11:59PM)

Overview

In this first milestone, your group will identify a real-world dataset suitable for a data warehousing project, build an automated ingestion pipeline that loads raw data into Hadoop HDFS, and produce a comprehensive data profiling report. Think of this milestone as your project blueprint. The quality issues and cleaning strategies you document here become the direct input for your ETL design in Assignment 3. Thoroughness now saves significant effort later.

Learning Outcomes

- Select and justify a dataset appropriate for data warehouse-scale distributed processing
- Configure a Hadoop HDFS environment and automate file ingestion using Python
- Systematically profile raw data to surface quality issues before any transformation begins

Task 1: Dataset Selection (5 marks)

Select a real-world dataset satisfying all of the following:

- Minimum **500,000 records**; large enough to justify distributed processing on HDFS and Spark
- At least **8 attributes** spanning multiple data types (numeric, categorical, and datetime)
- Clear **analytical value**; the data must be suitable for building a star or snowflake schema in A3
- From a credible source: Kaggle, UCI Machine Learning Repository, government open data portals, or equivalent

Deliverable: A concise dataset justification document (maximum 1 page) covering:

Element	Description
Dataset description	What the dataset contains, its origin, and a full source citation or URL
Warehouse suitability	Why it is well-suited for a data warehouse project
Business questions	At least 5 concrete business questions this dataset could help answer

Task 2: HDFS Ingestion Pipeline (8 marks)

Write a fully automated Python script (ingest.py) that executes the following steps in sequence without manual intervention:

Step	Action
Load	Loads or downloads the raw dataset from its source
Validate	Performs pre-upload validation: file integrity (size, format, extension), encoding detection (UTF-8, ISO-8859-1, etc.), row count verification
Upload	Uploads the validated dataset to HDFS using the <code>hdfs</code> Python library or Hadoop shell commands via <code>subprocess</code>
Organize	Organizes uploaded files in a structured HDFS directory, for example: <code>/warehouse/raw/dataset_name/year=2026/month=02/</code>
Log	Logs every step clearly using Python's <code>logging</code> module — successes, warnings, and errors must all be recorded



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Task 3: Data Profiling Report (17 marks)

Using Python (pandas + matplotlib/seaborn), produce a structured profiling report that will serve as the quality baseline for your A3 ETL pipeline. The report must include all of the following sections:

Section	Requirements
Schema Description	Column names, inferred data types, and representative sample values for each attribute
Missing Value Analysis	Count and percentage of nulls per column, visualized as a heatmap or bar chart
Statistical Summary	Mean, median, standard deviation, minimum, and maximum for all numeric columns
Distribution Analysis	Histogram or KDE plots for at least 5 key attributes with brief interpretation of each
Data Quality Issues	Explicit identification of duplicates, outliers, type inconsistencies, and formatting errors — each reported with a count or percentage
Proposed Cleaning Strategy	For every issue identified, state the exact cleaning action you will apply in A3 and justify why

Important: The cleaning strategy section carries significant weight. Vague statements such as *"we will handle missing values"* will receive no marks. Each identified issue must be paired with a specific, justified action.

Example of a strong cleaning strategy entry:

"Column fare_amount has 342 negative values (0.07% of records). These are data entry errors. Action: Replace negative values with the median fare amount for the same trip distance category, as median is robust against outliers in this skewed column."

Submission Requirements

Submit a single compressed archive containing:

File	Description
ingest.py	Fully automated Python ingestion script
hdfs_screenshot.png	Screenshot of HDFS showing uploaded files and directory structure
profiling_report.pdf	Data profiling report following the structure above
requirements.txt	All Python dependencies listed
README.md	GitHub documentation file, includes project overview, setup instructions, how to run ingest.py, group members, and submission details

File naming: GroupNumber_A2_BDA.zip

Advice to Students

Do not underestimate this milestone. The profiling report is a living technical document that will be referenced throughout A3 and A4; weak profiling leads to weak ETL design.

- **Set up your HDFS environment in the first two days** as configuration issues are the most common cause of late submissions
- Start the dataset search early and confirm your choice with the instructor before investing time in the pipeline
- Document everything as you go; a well-organized A2 makes A3 significantly easier